USAMA

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Informatics Engineering student at Tadulako University with strong interests in Embedded Systems, IoT, and Software Development. Passionate about continuous learning and adapting to emerging technologies. Experienced in both hardware and software development, including AI and web applications.

Technical skills

- Embedded System, IoT & Robotics
- PCB and electrical Schematics design
- C/C++ Language
- Python Language
- HTML & CSS (Tailwind)
- MySQL Database
- JS Language (React JS, Next JS, React Native, Node JS)
- TypeScript
- AI, Machine & Deep Learning modeling
- Visual Design & Multimedia

Tools

Arduino IDE, Eagle, EasyEda, Fritzing, Proteus, Wokwi, Thingspeak, Node-RED, Blynk, Thingsboard, Tinkercad, Google Collab, Visual Studio Code, Pycharm, Postman, Adobe Photoshop, Adobe Ilustrator, Adobe After Effects, Adobe Premiere, Coreldraw, Figma, Canva.

Soft Skills

- Analytical Thinking
- Teamwork
- Communication
- Problem solving
- Interpersonal Skills
- Leadership
- Adaptive
- Independent Work

Projects

Monitoring and controlling irrigation gates (ESP32, PHP, MySQL, JavaScript)

- Developed IoT system for real-time monitoring and remote control of irrigation gates
- Implemented ESP32-based data acquisition for water level and gate status
- Designed web dashboard for visualization and control interface
- Enabled efficient irrigation management by reducing manual monitoring

Euforia - PCB Defect Detection System (Python, Computer Vision, Deep Learning)

- Developed AI system for detecting PCB defects using image processing techniques
- Built a detection model to identify defective PCB boards

MP4 HD – Patient Monitor (Hardware Engineering, IoT)

- Developed a patient monitoring system for real-time health data observation
- Implemented hardware-based solution integrating sensors and microcontroller
- Completed as a final project for the MSIB6 Stechoq Hardware Engineering program

Landing Page – Pramuka SMANSA Poso (React.js, Tailwind)

- Developed responsive landing page using React.js
- Applied modern UI/UX principles for better user experience
- Contributed as a voluntary project for the organization

Plant Monitoring System (IoT, ESP32, Sensors, Blynk)

- Developed an IoT-based system for real-time plant monitoring
- Integrated sensors with ESP32 for environmental data collection
- Utilized Blynk platform to build a real-time monitoring dashboard
- Delivered a reliable solution to support plant research and client requirements

Mobile Monitoring Application (React Native)

- Developed a mobile application for monitoring and controlling environmental parameters
- Implemented features to track temperature, humidity, soil pH, water quality, and lighting systems
- Designed user-friendly interface for real-time monitoring and control
- Delivered solution tailored to meet client requirements

Asset Inventory and Management Portal (React.js, TypeScript, Express.js, MySQL)

- Developed a fullstack web-based asset management system
- Built frontend using React.js and TypeScript for responsive and scalable UI
- Developed RESTful API using Express.js for data processing and integration
- Implemented asset inventory, tracking, and management features
- Improved transparency and efficiency in municipal asset administration
- Delivered solution for Palu City Environmental Department

Experiences

05 November 2022

Jury in the Robotics Event PERMIKOMNAS REGION X GORONTALO

Hobby and Activities

Traveling, study.

Education Level

2020 - Present

Universitas Tadulako - Palu, Central Sulawesi

Bachelor Degree in Informatics Engineering, 3.66/4.00

August - December 2023

Study Independent Artificial Intelligence at Startup Campus

- AI Foundation
- Python Foundation
- Data Science
- Machine Learning and Deep Learning
- Computer Vision
- Git Collaboration

February - June 2024

Study Independent Hardware Engineering at PT. Stechoq

- Internet of Things
- Sensors dan Actuators
- Programmable Logic Controller (PLC) Supervisiory Control and Data Acquistion (SCADA)
- RFID Technology, Barcode and QR Code
- Industrial Control System
- Electronic Design
- Microcontroller

- Wireless Comunication Technology
- Analog and digital Electronic